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1. Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
int i, j, k, a, b, u, v, n, ne = 1;
int min, mincost = 0, cost[9][9], parent[9];
int find(int);
int uni(int, int);
void main()
{
printf("\n\tImplementation of Kruskal's Algorithm\n");
printf("\nEnter the no. of vertices:");
scanf("%d", &n);
printf("\nEnter the cost adjacency matrix:\n");
for (i = 1; i <= n; i++)
{
for (j = 1; j <= n; j++)
{
scanf("%d", &cost[i][j]);
if (cost[i][j] == 0)
cost[i][j] = 999;
}
}
printf("\nThe edges of Minimum Cost Spanning Tree are\n");
while (ne < n) {
for (i = 1, min = 999; i <= n; i++)
{
for (j = 1; j <= n; j++)
{
if (cost[i][j] < min)
{
min = cost[i][j];
a = u = i;
b = v = j;
}
}
}
u = find(u);
```



```
v = find(v);
if (uni(u, v))
{
    printf("%d edge (%d,%d) = %d\n", ne++, a, b, min);
    mincost += min;
}
cost[a][b] = cost[b][a] = 999;
}
printf("\n\tMinimum cost = %d\n", mincost);
}
int find(int i)
{
    while (parent[i])
        i = parent[i];
    return i;
}
int uni(int i, int j)
{
    if (i != j)
    {
        parent[j] = i;
        return 1;
    }
    return 0;
}
```

OUTPUT

```
cse@csat01:~/Desktop$ ./a.out

Implementation of Kruskal's algorithm
Enter the number of vertices:4

Enter the cost adjacency matrix:
0 3 4 1
3 0 1 0
4 1 0 0
3 0 2 0

The edges of minimum cost spanning tree are:
1edge(2,3)=1
2edge(4,3)=2
3edge(1,4)=3

Minimum cost=6
```